

RetailPulse – AI-Powered Customer Analytics & Demand Forecasting
Transforming Retail Intelligence with AI-Driven Insights

Zidio Internship Project

Developed By:

NAME

Project Duration:

Project Overview

RetailPulse is an AI-powered retail analytics platform designed to help businesses understand customer behavior, forecast product demand, and optimize inventory management using Machine Learning and MLOps practices. The project focuses on improving business intelligence through predictive analytics and real-time visual dashboards.

Vision & Objectives:

- Predict future product demand using AI forecasting models.
- Analyze customer purchasing behavior.
- Identify churn-prone customers.
- Improve inventory planning and reduce overstocking.

Target Users / Use Cases:

- Retail store managers
- E-commerce businesses
- Supply chain teams
- Business analysts and decision-makers

Business Value Delivered:

- Better forecasting accuracy
- Reduced inventory wastage
- Improved customer retention
- Data-driven decision making

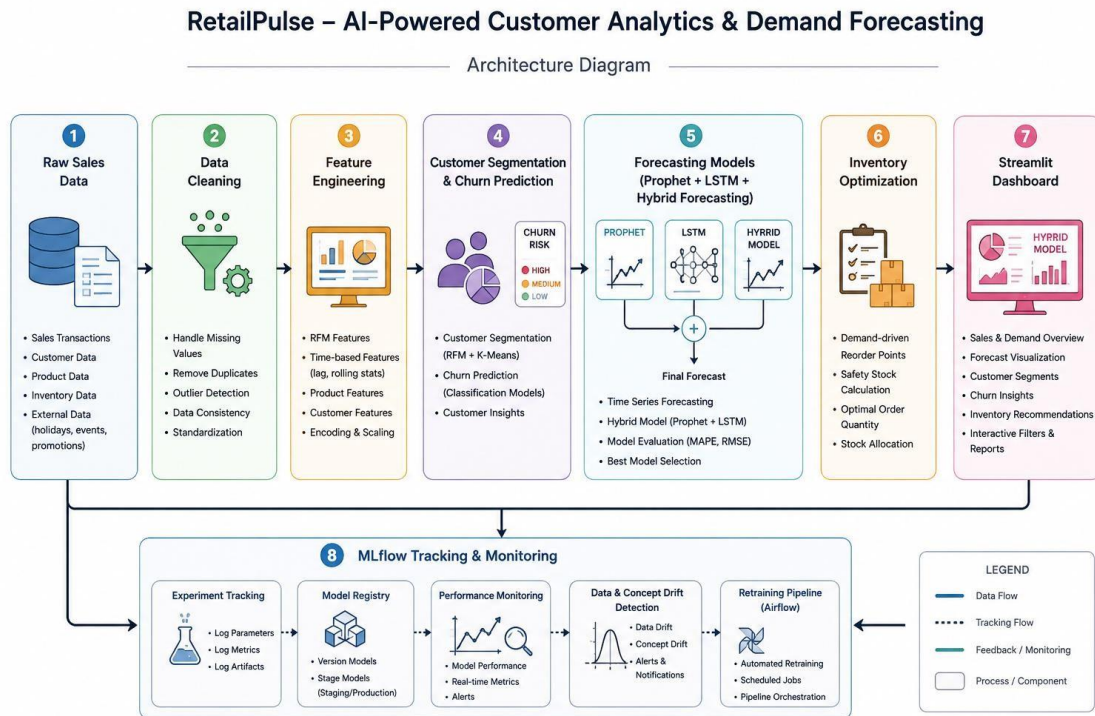
Non-functional Goals:

- Scalable ML pipeline
- Low latency dashboard visualization
- Reliable model tracking and monitoring
- Maintainable and modular architecture

Architecture Overview

RetailPulse follows an end-to-end AI and MLOps workflow:

Raw Sales Data → Data Cleaning → Feature Engineering → Customer Segmentation & Churn Prediction → Forecasting Models (Prophet + LSTM + Hybrid Forecasting) → Inventory Optimization → Streamlit Dashboard → MLflow Tracking & Monitoring.



The system is modular, scalable, and follows MLOps practices such as experiment tracking, model monitoring, retraining readiness, and workflow automation.

Execution Timeline

Week 1 (24 April – 30 April 2026)

- Requirement gathering and project planning
- Dataset collection and preprocessing
- Exploratory Data Analysis (EDA)

Week 2 (1 May – 7 May 2026)

- Feature engineering
- Customer segmentation using RFM analysis
- Churn prediction model implementation

Week 3 (8 May – 15 May 2026)

- Demand forecasting using Prophet and LSTM
- Hybrid forecasting model integration
- Inventory optimization model

Week 4 (16 May – 24 May 2026)

- Streamlit dashboard development
- Docker containerization
- Deployment and testing
- Final report preparation

Technical Highlights

Modeling Techniques:

- Customer Segmentation using clustering and RFM analysis
- Churn Prediction using Machine Learning
- Demand Forecasting using Prophet and LSTM
- Hybrid forecasting for improved prediction quality

Feature Engineering:

- Customer purchase frequency
- Recency and monetary value
- Sales trend extraction
- Time-series transformations

MLOps Practices:

- MLflow for experiment tracking
- Optuna for hyperparameter tuning
- Data drift detection
- Airflow-based pipeline orchestration

Challenges Faced:

- Handling inconsistent sales data
- Integrating multiple forecasting approaches
- Deployment configuration issues
- Docker container optimization

Solutions:

- Data preprocessing pipeline
- Hybrid forecasting approach
- Modular architecture
- Automated workflows

Deployment & Operations

Deployment Platform:

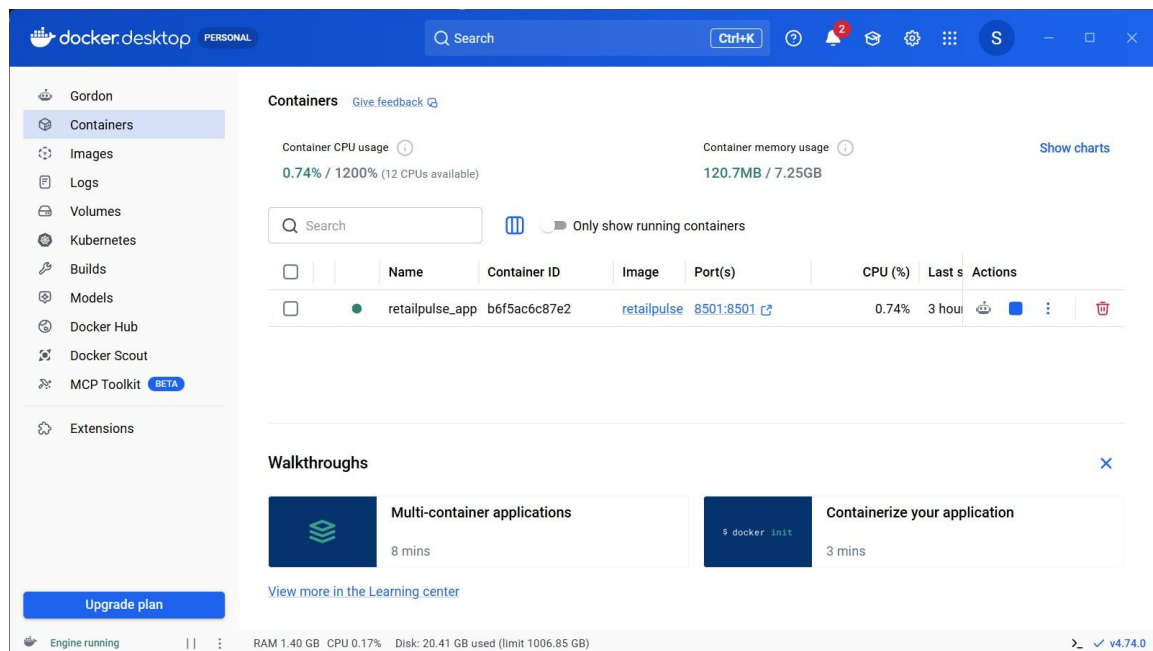
- Streamlit Cloud for web application hosting
- Docker for containerized deployment
- GitHub for version control

CI/CD Summary:

- Code managed through GitHub repository
- Containerized execution using Docker
- Continuous updates and deployment support

Monitoring Setup:

- Data drift detection mechanisms
- Model performance tracking
- MLflow experiment monitoring



[RetailPulse Dashboard · Streamlit](#) :- application deploy link

Personal Reflection

This project provided practical exposure to real-world Machine Learning and MLOps concepts. During development, we learned how to design scalable data pipelines, build predictive models, and deploy applications professionally.

Key Learnings:

- Hands-on experience with ML workflows
- End-to-end deployment process
- Dashboard development using Streamlit
- Practical implementation of forecasting and churn analysis

Industry Best Practices Applied:

- Modular code architecture
- Version control with GitHub
- Experiment tracking using MLflow
- Dockerized deployment

Future Roadmap Ideas:

- Real-time analytics integration
- Cloud-native deployment
- Recommendation systems
- Enhanced automation and monitoring

Key Features

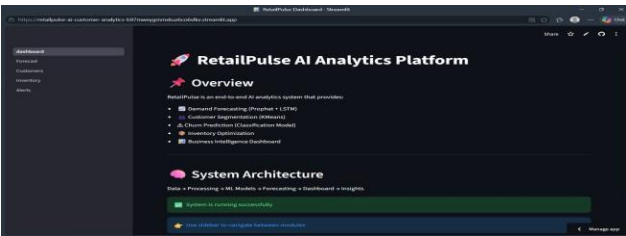
ID	Feature	Description	Acceptance Criteria
F1	Customer Segmentation	Groups customers based on behavior	Customers categorized successfully
F2	Churn Prediction	Predicts at-risk customers	Accurate churn insights generated
F3	Demand Forecasting	Predicts future demand	Reliable forecast trends shown
F4	Hybrid Forecasting	Combines multiple models	Improved prediction quality
F5	Inventory Optimization	Optimizes stock levels	Reduced overstocking risk
F6	Dashboard Analytics	Interactive visualization	Real-time insights accessible
F7	Drift Detection	Detects data changes	Alerts for model degradation
F8	MLflow Tracking	Tracks experiments	Performance history maintained

Technology Stack

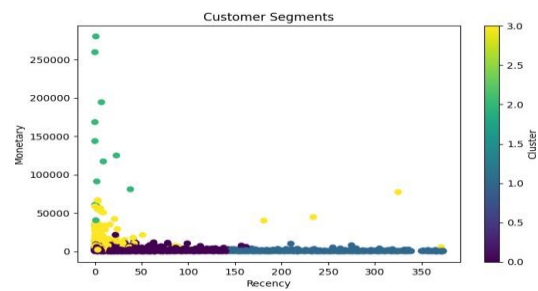
Category	Technology	Rationale / Alternatives
Programming	Python	Primary language for ML and analytics
Frontend	Streamlit	Interactive dashboard development
Machine Learning	Scikit-learn, TensorFlow	Model training and prediction
Forecasting	Prophet, LSTM	Time-series forecasting
MLOps	MLflow, Optuna, Airflow	Experiment tracking and orchestration
Containerization	Docker	Portable deployment
Version Control	Git & GitHub	Collaboration and code management

Visuals

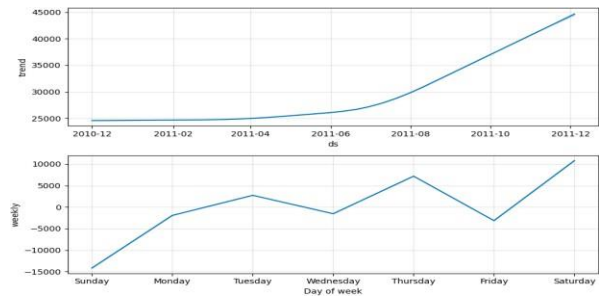
Dashboard Home Screen



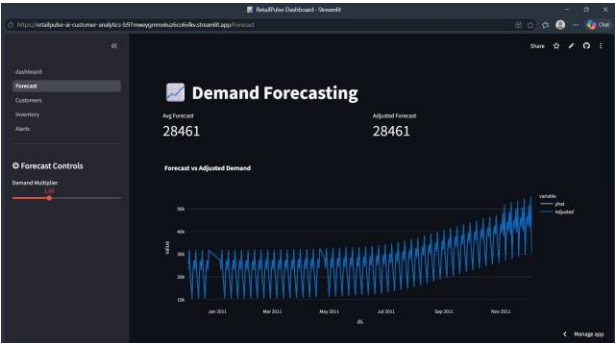
Customer Segmentation Output



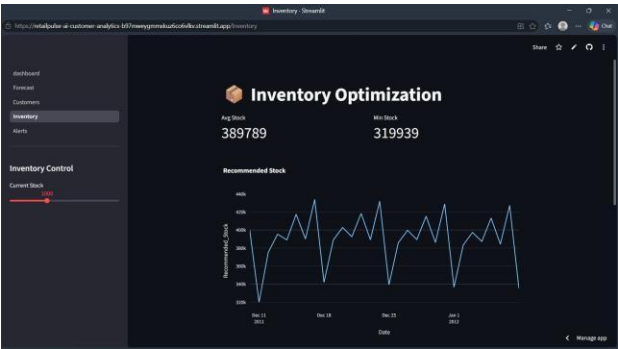
Forecast Graph



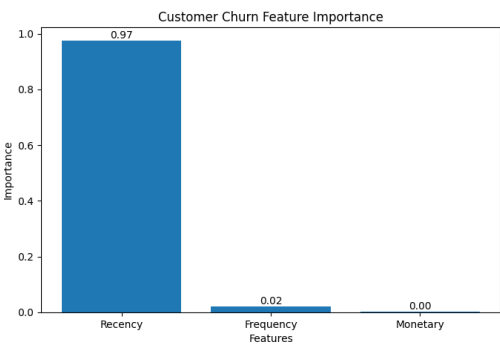
Forecast dashboard



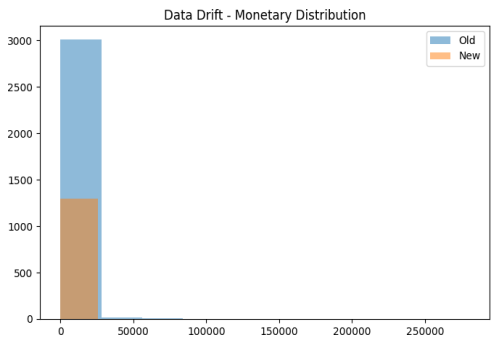
Inventory Dashboard



Churn feature importance



Drift Detection Result



OUTPUT:

CONCLUSION:

